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SEAT No. :

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**P41**

**APR - 18/TE/Insem. - 43**

**T.E. (Computer Engineering)**

**COMPUTER NETWORKS**

**(2012 Course) (Semester - II)**

*Time : 1 Hour]*

*[Max. Marks : 30*

*Instructions to the candidates:*

- 1) *Figures to the right indicate full marks.*
- 2) *Neat diagrams must be drawn wherever necessary.*
- 3) *Assume suitable data, if necessary.*

**Q1)** a) What is the difference between persistent & non persistent HTTP? Explain HTTP message format. [6]

b) Can we specify file transfer in a Web page? Explain with the help of suitable example. [4]

OR

**Q2)** a) Define Socket? Explain Socket primitives at client and server side for TCP communication with diagram. [7]

b) When web pages containing emails are sent out they are prefixed by MIME Header, Why? Explain MIME Header. [3]

**Q3)** a) What are four general techniques to improve quality of service? Explain any one in detail. [5]

b) Explain Three-Way Handshake Mechanism used by TCP for connection establishment. [2]

c) How does TCP achieves reliability? [3]

OR

**Q4)** a) What is the silly window syndrome problem? Suggest solution to overcome the problem. [6]

b) Explain TCP header format. [4]

**P.T.O.**

**Q5) a)** What are the various address Translation schemes used? Explain ARP message format? [6]

b) Explain distance vector routing algorithm. [4]

OR

**Q6) a)** Consider sending a 3000-byte datagram into a link that has an MTU of 500 bytes. Suppose the original datagram is stamped with the identification number 422. How many fragments are generated? What are their characteristics? [4]

b) Differentiate between IPV4 and IPV6 with respect to header format. [6]

